FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2021

TEST DATE: 19-12-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

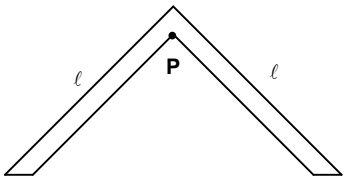
PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The half lives of a radioactive sample are 30 years and 60 years for α-emission and β-emission respectively. If the sample decays both by α-emission and β-emission simultaneously, the time after which, only one fourth of the sample remain is
   (A) 10 years
   (B) 20 years
   (C) 40 years
   (D) 45 years

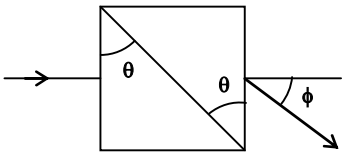
2. A system of two identical rods (L shaped) of mass m and length ℓ are resting on a peg P as shown in figure. If the system is displaced in its plane by a small angle θ, find the period of oscillation



   (A) $2\pi\sqrt{\frac{2\ell}{3g}}$
   (B) $2\pi\sqrt{\frac{2\sqrt{2}\ell}{3g}}$
   (C) $3\pi\sqrt{\frac{\sqrt{2}\ell}{3g}}$
   (D) $3\pi\sqrt{\frac{\ell}{3g}}$

3. In a vernier calipers 5 vernier scale divisions are equal to 4 main scale divisions. One main scale division is equal to 5 mm. A screw gauge with pitch equal to least count of above veriner callpiers is designed. Number of divisions on circular scale of this screw gauge is equal to 100. Least count of this screw is
   (A) 0.1 mm
   (B) 0.01 mm
   (C) 0.001 mm
   (D) 0.0001 mm

4. Two identical prisms with slightly different indices are located as shown. Angel θ is small. When a laser beam strikes one of the prisms perpendicular to the surface, the refracted ray is deviated by a small angle ϕ. The difference between the indices of refraction of the prism in terms of θ and ϕ is



   (A) $\frac{\phi}{\theta}$
   (B) $\frac{\theta}{\phi}$
   (C) $\frac{2\theta}{\phi}$
   (D) $\frac{2\phi}{\theta}$

5. The electric field in an electromagnetic wave is given by

$E = \left(50\ NC^{-1}\right)\sin\omega\left(t - \frac{x}{c}\right)$

Find the intensity of the wave ($c = 3 \times 10^8$ m/s)

(A) 3.3 W/m^2
(B) 0.33 W/m^2
(C) 33 W/m^2
(D) 330 W/m^2

6. A uniform rope having linear density λ hangs vertically from the ceiling and its lower end is free. A disturbance produced at the free end has a speed v_0 at point P midway on the rope. Then, the time taken by the disturbance pulse to reach the ceiling is

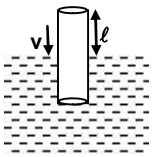
(A) $\frac{4v_0}{g}$

(B) $\frac{2\sqrt{2}v_0}{g}$

(C) $\frac{2\sqrt{2}v_0^2}{\lambda g}$

(D) $\frac{2\sqrt{2}v_0}{\lambda g}$

7. An open pipe of sufficient length is dipping in water with speed v vertically. If at any instant ℓ is length of tube above water, then the rate at which fundamental frequency of pipe changes is (c is the speed of sound in air)



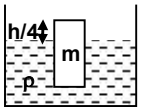
(A) $\frac{cv}{2\ell^2}$

(B) $\frac{cv}{4\ell^2}$

(C) $\frac{c}{2v^2t^2}$

(D) $\frac{c}{4v^2t^2}$

8. A solid cylinder of height h and mass m floats in a liquid of density ρ as shown in figure. Now the cylinder is released inside a liquid of density $\frac{\rho}{4}$, contained in a downward accelerated vessel. Determine the magnitude of acceleration of vessel A, for which cylinder sinks with relative acceleration $\frac{A}{3}$ with respect to vessel. Neglect any dissipative force



(A) $\frac{2}{3}g$

(B) $\frac{4}{3}g$

(C) $\frac{3}{4}g$

(D) $\frac{1}{3}g$

9. What is the radius of the circular orbit of a stationary satellite which remains motionless with respect to earth's surface? (T = 24 hours)

(A) $\left(\frac{gR^2T^2}{4\pi^2}\right)^{1/3}$

(B) $\left(\frac{gR}{4\pi^2T^2}\right)^{1/2}$

(C) $\left(\frac{gRT}{4\pi}\right)^{1/3}$

(D) $\left(\frac{gR}{2\pi T}\right)^{1/2}$

10. In YDSE of equal width slits, if intensity at the centre of screen is I_0, then intensity at a distance of $\beta/4$ from the central maxima is (β is the fringe width)

(A) I_0

(B) $\frac{I_0}{2}$

(C) $\frac{I_0}{4}$

(D) $\frac{I_0}{3}$

11. Two vehicles are horning at $\frac{320}{3}$Hz . They are moving directly towards each other, A is moving at 10 m/s and B is moving such that he hears a beat frequency of 5 Hz at t =0. If they are 90 m apart at t = 0, at what time (in sec) vehicles cross each other? (speed of sound = 330 m/s)

(A) 2 sec
(B) 4 sec
(C) 6 sec
(D) 8 sec

12. If the frequency of incident radiation is doubled in a photoelectric effect experiment, the kinetic energy of the fastest photo electron becomes

(A) double
(B) more than double
(C) less than double
(D) none of these

13. A parallel beam of fast moving electrons is incident normally on a narrow slit. A screen is placed at a large distance from the slit. If the speed of the electrons is increased, which of the following statement is correct ?

(A) Diffraction pattern is not observed on the screen in the case of electrons.
(B) The angular width of the central maximum of the diffraction pattern will increase.
(C) The angular width of the central maximum of the diffraction pattern will decrease.
(D) The angular width of the central maximum of the diffraction pattern will remain the same.

14. A small ball is projected with initial speed u and at an angle θ with horizontal from ground. The de-Broglie wavelength of ball at the moment its velocity vector becomes perpendicular to initial velocity vector is

(A) $\frac{h}{mu}$

(B) $\frac{h}{mu\sin\theta}$

(C) $\frac{h}{mu}\tan\theta$

(D) $\frac{h}{mu\cos\theta}$

15. The electric potential between a proton and an electron is given by $V = V_0 \log_e\left(\frac{r}{r_0}\right)$, where r_0 is a constant. Assuming Bohr's model to be applicable, write variation of r_n with n, where n being the principal quantum number

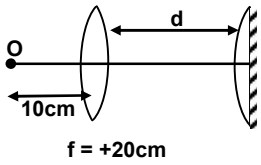
(A) $r_n \propto n$

(B) $r_n \propto \frac{1}{n}$

(C) $r_n \propto n^2$

(D) $r_\alpha \propto \frac{1}{n^2}$

16. A convex lens of focal length 20 cm and another planoconvex lens of focal length 40 cm are placed co-axially as shown in figure. The plano-convex lens is silvered on plane surface. What should be the distance d (in cm) so that final image of the object 'O' is formed on O itself



(A) 10
(B) 15
(C) 20
(D) 25

17. Vertical displacement of a plank with a body of mass m on it is varying according to law $y = \sin\omega t + \sqrt{3}\cos\omega t$, where y is in meter and t is in sec.
The minimum value of ω for which the mass just breaks off the plank and the moment it occurs first after t = 0 are given by (y is positive vertically upwards and g is gravitational acceleration)

(A) $\sqrt{\frac{g}{2}}$ rad / s, $\frac{\sqrt{2}}{6}\frac{\pi}{\sqrt{g}}$ sec

(B) $\frac{g}{\sqrt{2}}$ rad / s, $\frac{2}{3}\sqrt{\frac{\pi}{g}}$ sec

(C) $\sqrt{\frac{g}{2}}$ rad / s, $\frac{\pi}{3}\sqrt{\frac{2}{g}}$ sec

(D) $\sqrt{2g}$ rad / s, $\sqrt{\frac{2\pi}{3g}}$ sec

18. A ball of mass m and radius r is gently released in a viscous liquid. The mass of the liquid displaced by it is m′ such that m > m′. The terminal velocity is proportional to

(A) $\frac{m-m'}{r}$

(B) $\frac{m+m'}{r}$

(C) $\frac{m+m'}{r^2}$

(D) $(m-m')r^2$

19. An organ pipe closed at one end and open at another end has length 42 cm and diameter 10 cm. The air column in pipe vibrates in its second overtone with the maximum amplitude ΔP_0. The pressure amplitude at a distance 24 cm from the open end of the tube is

(A) $\frac{\Delta P_0}{2}$

(B) $\frac{\Delta P_0}{\sqrt{2}}$

(C) ΔP_0

(D) $\frac{\sqrt{3}\Delta P_0}{2}$

20. When the voltage applied to an X-ray tube increases from V_1 = 10 kV to V_2 = 20 kV, the wavelength interval between K_α line and cutoff wavelength of continuous spectrum increases by a factor 3. Atomic number of the metallic target is

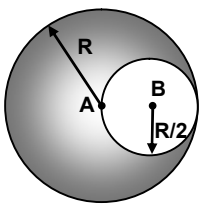
(A) 25
(B) 29
(C) 65
(D) 66

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A distant hot air balloon subtends 0.25° at the objective lens of a telescope. The image subtends 1.5° at eyepiece when viewed with relaxed eye. The objective and eyepiece are in a 35 cm long tube. Find the focal length of eyepiece lens (in cm).

22. Inside an insulated fixed share of radius R and uniform density ρ, there is a spherical cavity of radius $\frac{R}{2}$ such that the surface of the cavity passes through the centre of the sphere as shown in the figure. A particle of mass m_0 is released from rest at centre B of the cavity. Calculate velocity with which particle strike the center A of the sphere. If your answer is $\sqrt{\frac{P}{Q}\pi G\rho R^2}$, find the value of P + Q. (neglect earth's gravity)

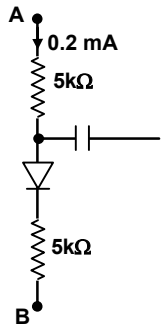


SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. An experiment is performed to obtain the value of acceleration due to gravity g by using a simple pendulum of length L. In this experiment time for 100 oscillations is measured by using a watch of 1 second least count and the value is 90.0 seconds. The length L is measured by using a meter scale of least count 1 mm and the value is 20.0 cm. Find the error (in %) in the determination of g.

24. In the given circuit, if the diode forward voltage drop is 0.3 V, Find the voltage difference (in volt) between A and B.



25. Due to a point isotropic sonic source, loudness at a point is L = 60 dB. If density of air is $\left(\frac{15}{11}\right)$ kg/m^3 and velocity of sound in air is v = 330 m/s, the pressure oscillation amplitude at the point of observation is (in N/m^2)

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Which of the following is not a detergent?
(A) Cetyltrimethylammonium bromide
(B) Sodium dodecylbenzene sulphonate
(C) Sodium lauryl sulphate
(D) Sodium stearate

27. Identify the correct statement from the following:
(A) Vapour phase refining is carried out for nickel by Van Arkel method
(B) Pig iron can be moulded into a variety of shapes
(C) Wrought iron is impure iron with 4% C
(D) Blister copper has blistered appearance due to evolution of CO_2

28. Measuring zeta potential is useful in determining which property of colloidal solution?
(A) Stability of the colloidal particles
(B) Size of the colloidal particles
(C) Viscosity
(D) Solubility

29. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds?
(A) $F^- < SCN^- < C_2O_4^{-2} < CN^-$
(B) $CN^- < C_2O_4^{-2} < SCN^- < F^-$
(C) $SCN^- < F^- < C_2O_4^{-2} < CN^-$
(D) $SCN^- < F^- < CN^- < C_2O_4^{-2}$

30. The mixture which shows positive deviation from Raoult's law is
(A) Acetone + Chloroform
(B) Chloroethane + Bromoethane
(C) Ethanol + Acetone
(D) Benzene + Toluene

31. The correct option for free expansion of an ideal gas under adiabatic condition is
(A) $q < 0, \Delta T = 0, w = 0$ and $\Delta S_{system} = 0$
(B) $q = 0, \Delta T = 0, w = 0$ and $\Delta S_{system} = 0$
(C) $q = 0, \Delta T = 0, w = 0$ and $\Delta S_{system} \neq 0$
(D) $q = 0, \Delta T < 0, w > 0$ and $\Delta S_{system} \neq 0$

32. The percentage composition by weight of an aqueous solution of a non-electrolyte solute (Molar mass = 150) which boils at 373.52 K is: $\left(\text{Given } K_b \text{ for } H_2O = 0.52 \text{ K kg mol}^{-1}\right)$
(A) 20%
(B) 13%
(C) 7%
(D) 25%

33. Which of the following statements regarding sulphur is not correct?
(A) Both rhombic and monoclinic sulphur are soluble in CS_2
(B) Both rhombic and monoclinic sulphur have S_8 molecules
(C) The stable form at room temperature is rhombic sulphur
(D) Monoclinic sulphur transforms to rhombic form when heated above 369 K

34. Which of the following statement is not correct?
(A) Catalyst is not involved in the reaction
(B) NO acts as homogeneous catalyst in the oxidation of SO_2 into SO_3
(C) Positive catalyst operates by providing alternate path for the reaction that involves a lower energy of activation
(D) Catalyst does not affect ΔG of the reaction

35. Find vapour pressure of 10 M $C_2H_5OH(aq)$ solution at 300 K. (Assume ideal solution)
Given
$d_{solution} = 1\ g/mL$
$P^o_{C_2H_5OH} = 40$ mm of Hg
$P^o_{H_2O} = 20$ mm of Hg
(A) 25 mm of Hg
(B) 30 mm of Hg
(C) 27.5 mm of Hg
(D) 32.5 mm of Hg

36. In the metallurgy of Cu, the material obtained from bottom of blast furnace after smelting, which is further subjected for bessemerization, is:
(A) Slag
(B) Impure Cu
(C) Matte
(D) Cuprous oxide

37. 25 mL sample of ozonized oxygen at NTP was passed through a solution of KI. The liberated iodine required 15 mL of 0.08 N $Na_2S_2O_3$ solution for complete titration. Calculate the volume of ozone at NTP in the given sample.
(A) 15.44 mL
(B) 13.44 mL
(C) 14.44 mL
(D) 5 mL

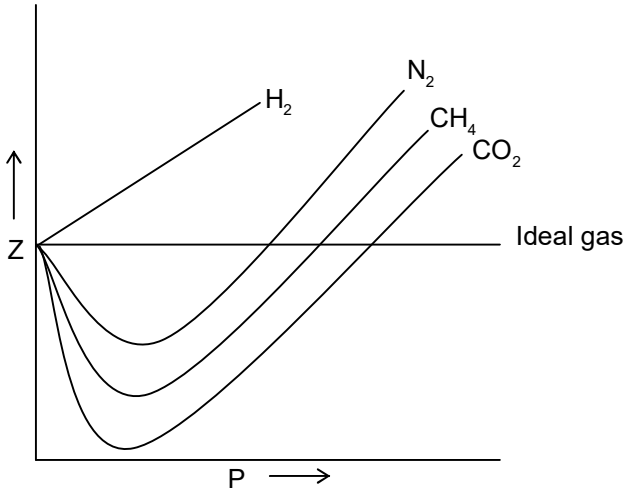
38. The total number of isomers for the complex $\left[Co(NO_2)_6\right]^{-3}$ will be
(A) 8
(B) 10
(C) 1
(D) 2

39. Calculate the standard voltage that can be obtained from ethane-oxygen fuel cell at $25^\circ C$
$C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell), \qquad \Delta G^\circ = -1229.41\ kJ/mol$
(A) + 0.91 V
(B) + 0.54 V
(C) + 0.72 V
(D) + 1.08 V

40. For the reaction
$2O_3(g) \longrightarrow 3O_2(g)$, $\Delta H, \Delta S$ and ΔG are respectively
(A) +ve, +ve, +ve
(B) –ve, +ve, –ve
(C) –ve, –ve, +ve
(D) +ve, –ve, –ve

41. A certain gas effuses out of two different vessels A and B. A has a circular orifice while B has a square orifice of length equal to the radius of the orifice of vessel A. The ratio of rate of effusion of the gas from vessel A to that from vessel B is (Assuming all other conditions to be same):
(A) $\pi : 1$
(B) $1 : \pi$
(C) 1 : 1
(D) 3 : 2

42. Compressibility factor $\left(Z = \frac{PV}{nRT}\right)$ is plotted against pressure



What is the correct order of liquefiability of the gases shown in the above graph?
(A) $H_2 < N_2 < CH_4 < CO_2$
(B) $CO_2 < CH_4 < N_2 < H_2$
(C) $H_2 < CH_4 < N_2 < CO_2$
(D) $CH_4 < H_2 < N_2 < CO_2$

43. An ideal mixture of liquids A and B with 2 moles of A and 2 moles of B has a total vapour pressure of 1 atm at a certain temperature. Another mixture with 1 mole of A and 3 moles of B has a vapour pressure greater than 1 atm. But if 4 moles of liquid C are added to the second mixture, the vapour pressure comes down to 1 atm. Given vapour pressure of C, $P_C^o = 0.8$ atm. Calculate the ratio of vapour pressures of pure A and pure B.
(A) 2
(B) $\frac{1}{2}$
(C) $\frac{7}{3}$
(D) $\frac{3}{7}$

44. Example of – vely charged colloid is
(A) As_2S_3
(B) $Fe(OH)_3$
(C) TiO_2
(D) Basic dyestuff

45. Which of the following is incorrect?
(A) $Fe^{+3} + K_4\left[Fe(CN)_6\right] \longrightarrow$ Blue colour
(B) $Fe^{+2} + K_3\left[Fe(CN)_6\right] \longrightarrow$ Blue colour
(C) $Fe^{+2} + K_4\left[Fe(CN)_6\right] \longrightarrow$ Blue colour
(D) $Cu^{+2} + K_4\left[Fe(CN)_6\right] \longrightarrow$ Red brown ppt.

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. How many of the following are oxide ores?
Chalcopyrites, Bauxite, Limonite, Siderite, Pyrolusite, Rutile, Cassiterite, Calamine, Cuprite, Chalcocite, Argentite, Cinnabar.

47. How many of the following species have at least one bond angle less than 100°?
H_2O, H_2S, ClF_3, IF_7, XeF_2, SF_6, $\left[TeF_5\right]^-$, XeF_4, O_3, SO_2, SO_3 .

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. A gas expands adiabatically reversibly such that $T \propto V^{-\frac{1}{2}}$. The value of $\gamma\left(C_{P,m} / C_{v,m}\right)$ will be

49. The standard free energy change for the following reaction is – 120 kJ/mol. What is the standard cell potential?
$2H_2O_2(aq.) \longrightarrow 2H_2O(\ell) + O_2(g)$

50. A 0.745% (w/V) KCl solution is ionized to the extent of 80%. The osmotic pressure at $27^\circ C$ of the solution will be (in atm): (Take R = 0.082 atm L K^{-1} mol^{-1}), (Atomic mass of K and Cl = 39 and 35.5 respectively)

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. Let ${}^nC_0 + {}^nC_3 + {}^nC_6 + ... = 341$ ($n \in N$), then the value of n can be
(A) 7
(B) 8
(C) 9
(D) 10

52. One of the roots of the equation $8x^2 - 65x + 8 = 0$ is
(A) $\left|\frac{1+\sin\frac{3\pi}{4}+i\cos\frac{3\pi}{4}}{1+\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}}\right|^3$
(B) $\left|\frac{1-\sin\frac{\pi}{4}+i\cos\frac{\pi}{4}}{1+\cos\frac{\pi}{4}-i\sin\frac{\pi}{4}}\right|^3$
(C) $\left|\frac{2+2\sin\frac{3\pi}{4}+2i\cos\frac{3\pi}{4}}{1-\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}}\right|^3$
(D) $\left|\frac{2+2\cos\frac{\pi}{4}-2i\sin\frac{\pi}{4}}{1+\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}}\right|^3$

53. There be two bags A and B each containing 5 tickets numbered 1, 2, 3, 4, 5. P_1 and P_2 be two persons who randomly pick one ticket from bags A and B respectively. The probability that P_1 has a number higher than P_2 is
(A) $\frac{1}{2}$
(B) $\frac{2}{5}$
(C) $\frac{3}{5}$
(D) $\frac{4}{5}$

54. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a 2×2 matrix such that trace of $A \cdot A^T$ is equal to 6 (where A^T is transpose matrix of matrix A), then maximum value of a^2bc^2d is equal to (a, b, c, d $\in$ R)

(A) 4
(B) 12
(C) 18
(D) 24

55. Let $f(x) = \begin{vmatrix} \operatorname{cosec} x & 2\cot x & \frac{e^x}{e^x - 1} \\ 3\sin x & \frac{1-\cos x}{x} & 2\tan x \\ 2 & 3x & 1 \end{vmatrix}$, then the value of $\lim\limits_{x \to 0} f(x)$ is equal to

(A) $\frac{3}{4}$
(B) 1
(C) $\frac{3}{2}$
(D) doesn't exist

56. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 4$ and $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = -10$, then the value of $|2\vec{a} + 3\vec{b} + 4\vec{c}|$ is equal to

(A) 4
(B) 8
(C) 12
(D) 24

57. If both roots of the equation $x^2 - (k + 1)x + k = 0$ lies between the roots of the equation $x^2 - 2kx + k^2 - 1 = 0$, then number of integral value(s) of k is/are

(A) 0
(B) 1
(C) 2
(D) infinite

58. The value of $\left|(10 + 24i)^{\frac{1}{2}} - (10 - 24i)^{\frac{1}{2}}\right|$ is equal to

(A) $2\sqrt{2}$
(B) $4\sqrt{2}$
(C) $6\sqrt{2}$
(D) $8\sqrt{2}$

59. Let A (α_1, β_1, γ_1) and B (α_2, β_2, γ_2) be two points on the lines $L_1 : \frac{x-1}{3} = \frac{y}{2} = \frac{z}{1}$ and $L_2 : \frac{x}{1} = \frac{y-1}{2} = \frac{z}{3}$. If distance AB is minimum, then the value of $(\alpha_1 + \beta_1 + \gamma_1) - (\alpha_2 + \beta_2 + \gamma_2)$ is equal to

(A) 0
(B) 1
(C) –1
(D) 2

60. If the value of $\sum_{r=2}^{10} \frac{r^2+r+1}{(r-1)(r)(r+1)(r+2)}$ is $\frac{k}{40}$, then k is equal to

(A) 29
(B) 27
(C) 23
(D) 21

61. Let α and β be the number of divisors of 88200 which are of the form (4k + 2) and (4k + 1) respectively, (k is integer). The value of $\alpha - \beta$ is equal to

(A) 8
(B) 12
(C) 24
(D) 36

62. Let k be the coefficient of x^{13} in the expansion of $(1 + 2x + 3x^2)^7$, then number of odd divisors of k is equal to

(A) 14
(B) 16
(C) 18
(D) 20

63. Six teams play a match with each other. The probability that neither a team win all the matches nor a team loose all the matches is

(A) $\frac{75}{216}$
(B) $\frac{57}{216}$
(C) $\frac{175}{216}$
(D) $\frac{157}{216}$

64. Let z be a complex number satisfying the $z^{117} = 1$, then minimum value of $|z - \omega|$ is equal to (ω is cube root of unity)

(A) $\frac{1}{2}$
(B) $\frac{1}{3}$
(C) 1
(D) none of these

65. The plane passing through (1, 2, –1) which is at farthest distance from (–1, 1, 2) passes through the point

(A) (0, 1, 2)
(B) (2, 0, –1)
(C) (1, 2, 3)
(D) (4, 1, 0)

66. Let x, y, $z \in R^+$ and x + y + z = 10, then number of ordered triplets of (x, y, z) such that $xy^2z^2 = 1000$ is
(A) 1
(B) 4
(C) 9
(D) none of these

67. Let P be a point with position vector $\vec{r}$ and A be a point with position vector $\vec{a}$. If $\vec{r}$ satisfy the equation $\left(\vec{r}-\left(\hat{i}+2\hat{j}-\hat{k}\right)\right)\cdot\vec{n}=0$ (where $\vec{n}$ is any vector in space) and $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$, then maximum value of $|\vec{r}-\vec{a}|$ is equal to
(A) 1
(B) $\sqrt{2}$
(C) $\sqrt{3}$
(D) $\sqrt{6}$

68. Let k be the number of ways of distribution of 5 identical red coloured balls and 7 identical blue coloured balls to 4 persons such that everyone gets atleast one ball, then the value of $\frac{k}{1990}$ is equal to
(A) 1
(B) 2
(C) 3
(D) 4

69. Let A be a 3×3 matrix and A^T is transpose matrix of A. If $A = 3A^T - 4I$ (when I is identity matrix of order 3), then |A| is equal to
(A) 2
(B) 5
(C) 8
(D) 10

70. Number of value(s) of 'a' for which the real roots of the equation $x^2 + (a^2 - 4)x + a^2 - 3a + 1 = 0$ are equal in magnitude and opposite in sign is/are
(A) 0
(B) 1
(C) 2
(D) 4

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. Let $\vec{a}=\hat{i}-2\hat{j}+2\hat{k}$ and $\vec{b}=2\hat{i}-\hat{j}-2\hat{k}$. $\vec{r}$ is a unit vector satisfying $\vec{r}\times\vec{a}=\vec{b}\times\vec{r}$, then the value of $\vec{r}\cdot\left(\hat{i}+\hat{j}+\hat{k}\right)$ is

72. Let total 5 letter words formed using the letters of the word EXAMINATION are k, then sum of digits of $\frac{k}{565}$ is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. Number of integral value(s) of x such that $x^2 + 14x + 53$ is perfect square is/are

74. Let sum of the first 13 terms of the A.P. $a_1, a_2, \ldots\ldots a_n$ ($n \in N$) be $52a_1$, then value of $\frac{a_{17}}{a_5}$ is equal to

75. Let perpendicular distance from (2, –1, 3) on the plane containing the lines $\frac{x-1}{3} = \frac{y}{2} = \frac{z}{1}$ and $\frac{x}{1} = \frac{y}{2} = \frac{z-1}{3}$ be k, then the value of k^2 is equal to